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Original Article

Integrative Image Processing Framework for Enhanced Detection of Leprosy-Associated Chronic Wounds

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Abstract

Automated leprosy chronic wound analysis from smartphone-acquired images remains hindered by uneven illumination, indistinct lesion margins, and poor spatial–textural integration. The CO-WinF framework introduces three specialized modules: AINCE, which performs tile-adaptive contrast remapping guided by local intensity distributions and edge-preserving smoothing to restore fine lesion textures; MEPS, which executes simultaneous multi-resolution encoding within a U-Net backbone enhanced by gradient-driven attention to emphasize boundary transitions and ensure accurate segmentation of irregular wound contours; and HSTFC, which leverages attention-weighted fusion of deep spatial embeddings and handcrafted LBP and HOG texture histograms, followed by classification via a gradient-boosted ensemble optimized for class imbalance. Validation on the CO2Wounds-V2 dataset yields 94.98% precision, 95.78% recall, 93.10% F1-score, and 87.09% IoU, surpassing existing state-of-the-art approaches. By integrating localized enhancement, edge-aware segmentation, and hybrid feature fusion in a computationally efficient pipeline, CO-WinF delivers robust, interpretable diagnostic support in resource-constrained clinical environments. Key novelties include the tile-adaptive remapping within AINCE, the gradient-driven attention integrated at scales in MEPS, and the attention-weighted fusion of spatial and textural features in HSTFC. By addressing preprocessing and feature-level integration challenges, CO-WinF establishes a benchmark for smartphone wound analysis.

Keywords

image processing, deep learning, u-Net, attention mechanism, leprosy, wound segmentation

Introduction

Although various preventive strategies have been implemented to curb the spread of leprosy, a total of 140,594 new cases were identified globally in 2021, corresponding to a detection rate of 17.83 per million individuals. Notably, during the same period, over 9000 new cases were documented among children aged 15 years or younger (WHO Global Leprosy Update, 2021).[1] India accounted for approximately 58% of the global leprosy burden, with paediatric leprosy prevalence rates varying widely across the country, ranging from 4% to 34%. In India, of a total of ~135,485 new cases of leprosy, 8.7% were among children.[2] These numbers reveal that a sizeable proportion of the newly detected cases occur in children. Leprosy incidence is highest among children aged 5–14 years, who account for approximately 6% of all reported cases.[3] The clinical manifestations of leprosy are largely determined by the host's immune response. According to the World Health Organization (WHO), leprosy is operationally classified into paucibacillary (PB) and multibacillary (MB) forms, based on the number of skin lesions—patients presenting with 1–5 lesions are categorized as PB, while those with more than 6 lesions are classified as MB.[4] In contrast, for research purposes, a more detailed classification is employed: the PB category encompasses tuberculoid (TT) and borderline tuberculoid (BT) forms, whereas the MB category includes borderline-borderline (BB), borderline-lepromatous (BL), and lepromatous (LL) forms of the disease.[5] Differential diagnosis for PB leprosy must consider other dermatological conditions such as pityriasis alba,

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syphilis, psoriasis, granuloma annulare, and sarcoidosis. In comparison, MB leprosy, particularly the LL form, requires distinction from systemic conditions including lymphoma, neurofibromatosis, and xanthoma.[6]

Previous studies have demonstrated that digital technologies can expedite leprosy diagnosis by enabling frontline health workers to transmit skin images via smart phones to specialized dermatologists for expert evaluation and referral.[7] The growing body of evidence supporting the application of machine learning for rapid and accurate image-based diagnosis in fields such as radiology, pathology, and dermatology prompted the development of an Artificial Intelligence (AI)-driven "diagnosis accelerator" for leprosy, termed AI4Leprosy. This system integrates skin images with clinical features and reported symptoms to enhance diagnostic precision.[8] Notably, much of the existing AI-driven diagnostic success in dermatology stems from melanoma research, where deep convolutional neural networks (CNNs) have achieved diagnostic accuracies comparable to those of experienced dermatologists in distinguishing malignant from benign lesions.[9] Deep neural networks have demonstrated superior performance compared to clinical specialists in differentiating melanoma from benign mimickers such as nevi and seborrheic keratoses in dermoscopic images. The integration of larger training datasets and continual advancements in algorithmic architectures has further enhanced diagnostic accuracy, particularly by augmenting clinical decision-making in cases where physicians report low diagnostic confidence.[10] Chronic wounds associated with leprosy present a distinct clinical challenge, arising from a multifactorial interplay of peripheral neuropathy, vascular compromise, and immune dysfunction, each contributing to the development of persistent, non-healing ulcers. Despite notable progress in chronic wound management and machine learning applications in other medical domains, the specific challenges posed by leprosy-related wounds remain insufficiently addressed. There is an urgent need for the development of robust, computationally efficient diagnostic and therapeutic frameworks capable of accurately segmenting, monitoring, and classifying these complex lesions. Advancing research in this area holds the potential to significantly improve early detection, personalized wound management, and ultimately mitigate the socio-economic impact and stigma associated with leprosy, particularly in resource-constrained regions where the disease remains endemic.

Literature Survey

Leprosy remains a significant global health concern, particularly among underserved populations. Although considerable progress toward elimination has been achieved, approximately 200,000 new cases are still reported each year. The persistent lack of rapid diagnostic tools contributes to delayed diagnosis, resulting in irreversible neurological damage, social stigma, and sustained disease transmission. To expedite diagnosis, Barbieri et al[11] developed an Artificial Intelligence (AI)-based diagnostic assistant, AI4Leprosy, utilizing skin images and clinical data. The dataset comprised 1229 skin images and 585 clinical metadata entries, collected from 222 patients at a Brazilian national leprosy referral centre. A convolutional neural network (CNN)-based AI algorithm was employed to assess diagnostic performance. Analysis revealed that key clinical indicators, including thermal sensitivity loss, nodules and papules, foot paraesthesia, number of lesions, gender, scaling surfaces, and pruritus, were significantly associated with leprosy diagnosis, with several features showing negative associations. The AI model demonstrated high classification performance, achieving an overall diagnostic accuracy of 90% and an area under the receiver operating characteristic curve (AUC) of 96.46%.

Beesetty et al[12] investigated the application of AI-driven image markers for early leprosy screening to improve case detection and reduce healthcare costs. Using a multimodal imaging protocol, they collected 368 clinically confirmed leprosy lesions and 28 non-leprosy skin lesions from 151 eligible patients. A Siamese network-based Few-Shot Learning (FSL) model was developed to address the limited dataset size and the small number of disease classes. The model, fine-tuned based on detailed skin morphological features, was evaluated using distinct training and testing sets. For two-way one-shot classification tasks, the model achieved an accuracy of 91.25% on the training set and 73.12% on the testing set. The highest sensitivity and specificity values for the test set were recorded at 72.39% and 73.66%, respectively.

The CO2Wounds-V2 dataset comprises 764 RGB wound images collected from leprosy patients for 1 year and 9 months in a developing country. This dataset provides a comprehensive benchmark for evaluating the performance of segmentation algorithms, capturing a diverse array of chronic wounds associated with leprosy. Its real-world applicability is further enhanced by the inclusion of images obtained using smartphone cameras, making it particularly relevant for practical deployment in resource-limited settings. Sanchez et al[13] anticipate that this dataset will be a valuable resource for the research community, fostering the development of novel algorithms and advancing the state-of-the-art in automated wound analysis.

Sulthana et al[14] propose an end-to-end framework for skin lesion classification, emphasizing two algorithms for image preprocessing: one for segmentation and another for feature extraction. The framework includes a deep convolutional neural network model, S-MobileNet, which classifies seven distinct skin lesion types. For preprocessing, the HAM10000 dataset, comprising 10,000 dermatoscopic images from diverse populations, was employed. Image preprocessing was carried out using exploratory data analysis (EDA), a modified Gaussian filtering algorithm, and spatial frequency texture analysis (SFTA) for segmentation and feature extraction. The S-MobileNet model, designed for efficiency, employs the ReLU and Mish activation functions and incorporates layer compression techniques at intermediary stages. The Mish activation function demonstrated superior performance over ReLU, while the compressed

S-MobileNet model outperformed the original S-MobileNet in terms of classification accuracy without compromising computational efficiency.

Yolcu et al[15] developed a quantitative, objective classification tool designed for dermatologists and researchers to classify seven types of skin lesions, including monkeypox. The system utilizes Vision Transformer models in conjunction with deep convolutional neural networks, trained via transfer learning and augmented with bagging ensemble learning techniques. The model achieved an accuracy of 81.91%, a Jaccard index of 65.94%, a precision of 87.16%, a recall of 74.12%, and an F-score of 78.16%, surpassing existing literature across multiple performance metrics.

Ji et al.[16] introduced a novel EFAM-Net model for skin lesion classification. It uses a new Attention Residual Learning ConvNeXt block to extract low-level features like colors and textures, replacing deep-layer blocks with Parallel ConvNeXt blocks for richer features. A Multi-scale Efficient Attention Feature Fusion block enhances feature extraction at various scales, capturing more comprehensive features in specific layers. Experimental evaluation on the ISIC 2019 and HAM10000 public and private datasets reveals that EFAM-Net achieves top classification performance, with overall accuracy of 92.30%, 93.95%, and 94.31%, respectively.

Remya et al[17] introduced a novel framework that integrates a Vision Transformer model with transfer learning, a channel attention mechanism, and region of interest (ROI) analysis for precise skin condition detection, including skin cancer. The approach utilizes a comprehensive dataset comprising macroscopic dermoscopic images and patient metadata. By incorporating these advanced techniques, the framework enhances key diagnostic parameters such as sensitivity, specificity, and precision, surpassing the performance of traditional methods. Achieving a remarkable accuracy of 99%, this method underscores the transformative potential of deep learning and multimodal data analysis in improving dermoscopic diagnosis.

However, the exiting approaches during the pre-processing stage inadequately addresses the inherent variability of smartphone-captured images. Inconsistent illumination, noise, and contrast discrepancies frequently lead to a loss of fine-grained lesion details, directly impacting subsequent segmentation performance. Also, segmentation techniques that predominantly rely on standard deep learning architectures struggle with heterogeneous wound textures and irregular, indistinct edges. This results in suboptimal lesion boundary delineation, adversely affecting the accuracy of both feature extraction and the overall lesion characterization. Moreover, the classification models employed to date tend to underutilize the spatial context and intricate textural nuances embedded within the wound images. Models that focus solely on classical texture descriptors fall short of achieving robust generalization, especially in imbalanced datasets. Addressing these challenges is critical not only for improving diagnostic accuracy but also for enhancing the computational efficiency and robustness of the entire workflow. By pinpointing these specific gaps, our research aims to develop a unified, computationally efficient framework that systematically mitigates these issues and elevates the overall performance of automated leprosy analysis.

Skin Lesion Vision Integrated Framework

The advent of open-access datasets, such as CO2Wounds-V2, has catalysed research in automated chronic wound analysis, particularly for patients with leprosy; yet, critical technical gaps persist in existing methodologies. This research proposes a novel Skin Lesion Vision Integrated Framework, designed to streamline and enhance the traditional pipeline while maintaining computational efficiency. CO-WinF comprises three core modules, each addressing one of the specific technical gaps: Adaptive Illumination Normalization and Contrast Enhancement (AINCE) for pre-processing, Multi-scale Edge-Preserving Segmentation (MEPS) for lesion delineation, and Hybrid Spatial-Texture Fusion Classifier (HSTFC) for robust classification, as shown in Figure 1.

The proposed framework thus ensures that pre-processing is tailored to enhance image quality, segmentation is robust to the irregularities of wound morphology, and classification fully leverages the multifaceted nature of the data. In comparison to traditional methods, the proposed framework is less complex computationally, exhibits improved accuracy and generalizability, and provides a more reliable, end-to-end solution for automated leprosy analysis using the CO2Wounds-V2 dataset. This integrated approach is poised to advance clinical decision-making and support scalable diagnostic applications in resource-constrained environments.

Adaptive Illumination Normalization and Contrast Enhancement (AINCE)

In the pre-processing stage, many studies inadequately address the inherent variability of smartphone-captured images. Inconsistent illumination, noise, and contrast discrepancies frequently lead to a loss of fine-grained lesion details, directly impacting subsequent segmentation performance which is tackled by the first module, Adaptive Illumination Normalization and Contrast Enhancement (AINCE). Unlike conventional global histogram equalization, AINCE operates on a localized scale by applying adaptive histogram equalization techniques that adjust contrast based on the local pixel intensity distribution. Additionally, it incorporates an edge-preserving noise reduction filter, leveraging bilateral filtering to smooth out noise without blurring critical lesion boundaries. This dual approach ensures that the pre-processed images retain essential high-frequency details and textures, establishing a robust foundation for the downstream segmentation task. By dynamically

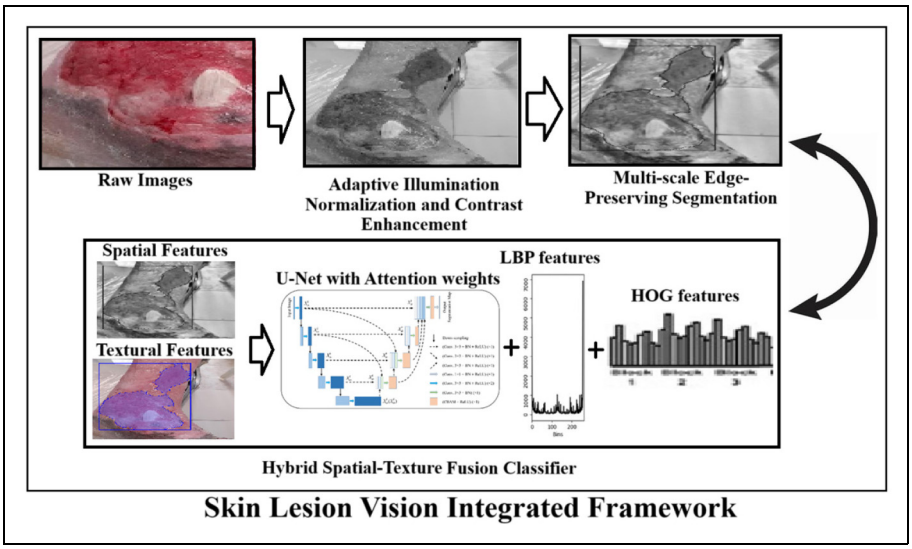


Figure 1. Proposed architecture.

normalizing image intensities and enhancing contrast in a context-sensitive manner, AINCE significantly improves the signal-to-noise ratio, thus addressing the first gap in a computationally efficient manner.

Adaptive Illumination Normalization and Contrast Enhancement (AINCE) divides each image into tiles and applies Contrast-Limited Adaptive Histogram Equalization (CLAHE) to redistribute pixel intensities.[18] CLAHE enforces a clip limit to prevent noise over-amplification and limit the dominance of certain histogram bins. AINCE then employs an edge-preserving bilateral filter to smooth noise while maintaining critical lesion boundaries. This combination of adaptive, region-based histogram equalization and bilateral noise reduction improves the signal-to-noise ratio of smartphone-captured wound images and establishes a strong foundation for downstream segmentation tasks.

First the input image $I(x, y)$ is partitioned into non-overlapping tiles of size $M \times N$. Within each tile T, we compute the local histogram $h_T(i)$, apply a clip limit α to obtain

$$h_T^c(i) = \min(h_T(i), \alpha) \tag{1}$$

redistribute the excess uniformly across all L gray levels, and form the clipped-and-redistributed histogram $h_T^r(i)$. The local cumulative distribution function is then

$$\mathrm{CDF}_T(k) = \sum_{i=0}^{k} \frac{h_T^r(i)}{|T|} \tag{2}$$

and each pixel intensity is remapped as

$$I'(x, y) = (L-1)\mathrm{CDF}_T(I(x, y)) \tag{3}$$

This contrast-limited adaptive histogram equalization step enhances local contrast without over-amplifying noise in homogeneous regions. To suppress residual noise while preserving critical lesion edges, AINCE then applies an edge-preserving bilateral filter to the CLAHE output I'. For each pixel p, the filtered intensity is

$$I''(p) = \frac{1}{W_p} \sum_{q \in \Omega} I'(q) g_s(\|p-q\|) g_r(|I'(q) - I'(p)|) \tag{4}$$

where g_s and g_r are Gaussian kernels controlling spatial proximity and radiometric similarity, respectively. By combining localized CLAHE with bilateral smoothing, AINCE dynamically normalizes uneven illumination, enhances fine-grained contrast, and preserves lesions establishing a robust, detail-rich foundation for subsequent segmentation. AINCE further refines the CLAHE-processed image $I'(x, y)$ by applying an edge-preserving bilateral filter,[19] which smooths noise while retaining lesion boundaries critical for downstream segmentation. The bilateral filter at the pixel p is defined as:

$$W_p = \sum_{q \in \Omega_p} G_{\sigma_s}(\|p-q\|) G_{\sigma_r}(|I'(p) - I'(q)|) \tag{5}$$

where

$$G_{\sigma_s}(d) = \exp\left(-\frac{d^2}{2\sigma_s^2}\right), \quad G_{\sigma_r}(r) = \exp\left(-\frac{r^2}{2\sigma_r^2}\right) \tag{6}$$

are the spatial and range Gaussian kernels, respectively. In our framework, the neighborhood Ω_p is chosen to match the CLAHE tile size (eg, $= 23\sigma_s + 1$), ensuring that smoothing adapts to the local scale of smartphone-captured wound textures. We set σ_s proportional to the tile dimensions to balance noise reduction against feature preservation, and compute σ_r adaptively as

$$\sigma_r = \beta \sigma_T \tag{7}$$

where σ_T is the standard deviation of intensities within the tile and β is a small constant (0.15) tuned on a validation subset. This adaptive bilateral filtering effectively suppresses sensor noise and illumination artifacts common in smartphone images of chronic wounds while maintaining high-frequency lesion edges for precise segmentation.

Adaptive Illumination Normalization and Contrast Enhancement (AINCE) combines localized contrast adjustment and edge-preserving noise reduction to produce consistently high-quality wound images from heterogeneous, smartphone-captured data. By partitioning each image into tiles and applying contrast-limited Adaptive Histogram Equalization, AINCE dynamically equalizes illumination and enhances fine details without amplifying noise. A subsequent, tile-aware bilateral filter then smooths residual sensor and lighting artifacts while faithfully preserving critical lesion boundaries. This two-stage process not only boosts the signal-to-noise ratio and maintains high-frequency textures essential for accurate segmentation, but it also operates with minimal computational overhead—making it suitable for real-time analysis on smartphone-acquired images . In practical terms, AINCE ensures that downstream segmentation and classification modules receive uniformly enhanced inputs, leading to more reliable lesion delineation, improved diagnostic accuracy, and robust performance even in resource-constrained or variable-lighting environments.

Multi-Scale Edge-Preserving Segmentation (MEPS)

The second module, Multi-scale Edge-Preserving Segmentation (MEPS), is designed to overcome the shortcomings of conventional segmentation methods that fail to accurately delineate the irregular and heterogeneous wound boundaries which results in suboptimal lesion boundary delineation, adversely affecting the accuracy of both feature extraction and lesion characterization. MEPS employs a multi-scale approach that processes the image at several resolutions simultaneously. At its core, it integrates a U-Net-based architecture augmented with an edge-aware attention mechanism, which refines feature maps by emphasizing strong gradient transitions indicative of lesion borders. Furthermore, MEPS employs a fusion of shallow features capturing fine textures and deeper features that encapsulate semantic context, thereby generating a more precise segmentation mask. MEPS is designed to robustly delineate lesion boundaries in chronic wound images by processing the image at multiple resolutions simultaneously. This mechanism dynamically emphasizes regions with strong gradient transitions as important cues for lesion boundaries by computing a spatial attention map.

In MEPS, we augment the standard U-Net encoder[20] with an explicit edge-extraction unit at each down-sampling level to ensure that boundary cues directly inform the decoder. Concretely, for the l_{th} encoder block, feature encoding is done in two successive 3×3 convolutions with ReLU activations produce the feature map F^l:

$$F^l = f(W_2^l * f(W_1^l * F^{l-1} + b_1^l) + b_2^l) \tag{8}$$

where $*$ is convolution, $W_{1,2}^l$ and $b_{1,2}^l$ are learnable weights and biases, and f is ReLU.

Edge map computation was then computed using an explicit edge response E^l by convolving F^l with Sobel-like gradient kernels S_x and S_y:

$$E^l(x,y) = \sqrt{(F^l * S_x)^2 + (F^l * S_y)^2} \quad \text{with} \quad S_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, S_y = S_x^T \tag{9}$$

which highlights horizontal and vertical intensity transitions corresponding to lesion borders. Also, an enhanced skip connection was given by the encoder feature, and its edge map is concatenated to form the enriched skip tensor:

$$C^l = \text{Concat}(F^l, E^l) \in \mathbb{R}^{H_l \times W_l \times 2C_l} \tag{10}$$

For instance, for a given encoder feature map $U(x, y)$, the attention map $A(x, y)$ maybe defined as:

$$A(x,y) = \frac{1}{1 + \exp(-\gamma(\|\nabla I(x,y)\| - \tau))} \tag{11}$$

where $\|\nabla I(x, y)\|$ denotes the gradient magnitude at pixel (x, y), γ controls the sensitivity to gradient differences, and τ sets a threshold for emphasizing edge information. The feature maps are then modulated by this attention map via element-wise multiplication, thus enhancing the representation of lesion borders.

Moreover, MEPS achieves improved segmentation precision by fusing features from different network depths. Shallow features, which capture fine texture details, are combined with deeper features that encode semantic context. This fusion is implemented through a concatenation followed by a 1×1 convolution:

$$F_{\text{fused}} = \sigma(W_{\text{fuse}} * \text{Concat}(U_{\text{shallow}}, U_{\text{deep}})) \tag{12}$$

where W_{fuse} represents the learned fusion weights and σ is a nonlinear activation function. This process creates a more discriminative feature representation that supports the generation of a precise segmentation mask.

In the corresponding decoder block, the up-sampled decoder feature D^{l+1} is merged with C^l and passed through a 3×3 convolution:

$$D^l = f(W_d * \text{Concat}(\text{Up}(D^{l+1}), C^l) + b_d) \tag{13}$$

where $\text{Up}(\cdot)$ is bilinear upsampling, and W_d, b_d are the decoder's learnable parameters.

By explicitly extracting and feeding edge maps into the skip pathways, MEPS's U-Net backbone preserves and emphasizes lesion boundary information crucial for accurate delineation of irregular, low-contrast wound edges.

Overall, MEPS offers a robust segmentation framework that not only captures the multi-scale and heterogeneous nature of chronic wound textures but also preserves critical edge details through an edge-aware attention strategy. The combination of these techniques leads to enhanced lesion delineation, ultimately providing a more reliable foundation for subsequent diagnostic and classification tasks in automated leprosy wound analysis.

Hybrid Spatial-Texture Fusion Classifier (HSTFC)

To bridge the gap in the final classification stage, the Hybrid Spatial-Texture Fusion Classifier (HSTFC) module fuses complementary information from both deep spatial features and handcrafted texture descriptors. Conventional classification models employed tend to underutilize the spatial context and intricate textural nuances embedded within the wound images. Models that focus solely on classical texture descriptors fall short of achieving robust generalization, especially in imbalanced datasets. To overcome this, HSTFC extracts deep features using a lightweight convolutional neural network (CNN) backbone optimized for efficiency while concurrently computing classical descriptors such as local binary patterns (LBP) and histograms of oriented gradients (HOG).[21]

The HSTFC module begins by extracting deep spatial features from a lightweight CNN backbone optimized for deployment. Each convolutional block l computes

$$F^l = f(W^l * F^{l-1} + b^l) \tag{14}$$

where W^l and b^l are the learnable kernel and bias, * denotes convolution, and $f(\cdot)$ is ReLU activation ($f(x) = \max(0, x)$). After several conv-ReLU layers, a max-pooling layer reduces spatial dimensions:

$$P^l(i, j, k) = \max_{(m,n)\in\mathcal{W}} F^l(i \cdot s + m, j \cdot s + n, k) \tag{15}$$

with pooling window $\mathcal{W}$ and stride s. Finally, a global average pooling produces a fixed-length deep feature vector

$$D_k = \frac{1}{HW}\sum_{i=1}^{H}\sum_{j=1}^{W} F^L(i, j, k) \tag{16}$$

where H, W are the final feature map's height and width.

Concurrently, handcrafted texture descriptors are computed on the pre-processed image I. Local Binary Patterns (LBP) generate a P -bit code per pixel by thresholding its P neighbors $\{I_p\}$ against the center intensity I_c :

$$\begin{aligned}\mathrm{LBP}(x_c, y_c) &= \sum_{p=0}^{P-1} s(I_p - I_c)2^p, \;\; s(z) \\ &= \begin{cases} 1, & z \geq 0 \\ 0, & z < 0 \end{cases},\end{aligned} \tag{17}$$

and aggregating these codes into a histogram H_{LBP} over the lesion region.

Histogram of Oriented Gradients (HOG) divides the image into cells, computes gradients

$$\begin{aligned}G_x &= I(x+1, y) - I(x-1, y), \;\; G_y \\ &= I(x, y+1) - I(x, y-1)\end{aligned} \tag{18}$$

then forms per-cell histograms of gradient orientations $\theta = \arctan2(G_y, G_x)$ weighted by magnitude $M = \sqrt{G_x^2 + G_y^2}$. Each cell's HOG descriptor H_{HOG} concatenates these histograms across orientation bins.

The deep feature vector $D \in \mathbb{R}^n$ and the texture vector $T = [H_{\mathrm{LBP}}, H_{\mathrm{HOG}}] \in \mathbb{R}^m$ are merged via an attention-based fusion:

Concatenation is formulated as,

$$F = \mathrm{Concat}(D, T) \in \mathbb{R}^{n+m} \tag{19}$$

Attention weights w are learned by

$$u = W_a F + b_a, \;\; w_i = \frac{\exp(u_i)}{\sum_j \exp(u_j)} \tag{20}$$

Producing $w \in \Delta^{n+m-1}$. Weighted fusion is given by,

$$F' = w \odot F \tag{21}$$

which dynamically emphasizes either spatial or textural cues based on lesion appearance.

Finally, F' feeds into a gradient-boosted tree ensemble (eg, XGBoost) for classification. At iteration m, a new tree $h_m(x)$ fits the pseudo-residuals of the current model $F_{m-1}(x)$, and the ensemble is updated as

$$F_m(x) = F_{m-1}(x) + \sum_{j=1}^{J_m} \gamma_{jm} 1_{R_{jm}}(x) \tag{22}$$

where each leaf R_{jm} contributes weight γ_{jm} chosen to minimize the loss L over samples in R_{jm}. Class imbalance is handled via weighted objective functions and scale-pos-weight hyperparameters. This hybrid approach leverages both deep spatial context and proven texture descriptors, yielding robust, interpretable, and generalizable classification for leprosy wound images. This hybrid approach not only enhances the overall predictive performance, yielding improvements in sensitivity and specificity but also provides a more interpretable decision-making process by correlating deep representations with well-understood texture features. By integrating spatial and textural information in a unified model, HSTFC offers a significant advantage over existing methods that rely on a single modality of feature extraction.

Result and Discussion

Implementation Setup

All experiments were conducted on a standard desktop workstation running Windows with an Intel Core i7-10700 K CPU

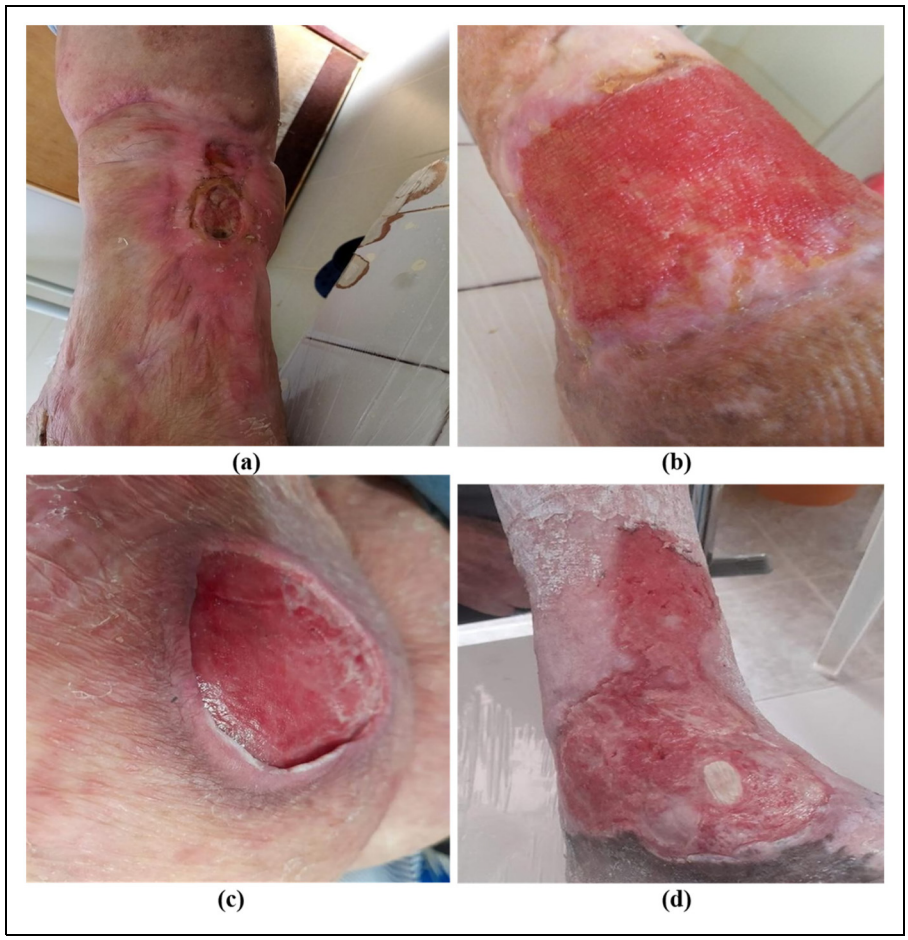


Figure 2. Sample images from the dataset.

(8 cores/16 threads at 3.8 GHz), and 32 GB DDR4 RAM. The framework was implemented in Python 3.9 within a Conda environment to ensure dependency isolation and reproducibility. Core libraries included PyTorch 1.10.0 with CUDA 11.1 and torchvision 0.11.0 for deep learning, NumPy 1.21.2 and Pandas 1.3.3 for data manipulation, scikit-learn 1.0 for evaluation metrics, OpenCV 4.5.3 and scikit-image 0.18.3 for image processing, Albumentations 1.0.3 for data augmentation, and XGBoost 1.4.2 for the ensemble classifier. A fixed random seed (42) was set for NumPy and PyTorch to guarantee reproducibility. Data were loaded via custom PyTorch Dataset classes with a batch size of 16 and four worker threads. Training utilized the Adam optimizer (initial learning rate 1×10^{-4}, weight decay 1×10^{-5}) alongside a ReduceLROnPlateau scheduler to adaptively lower the learning rate.

Dataset Description

This research utilized the CO2Wounds-V2 dataset, which is an open-access, high-quality resource designed to facilitate the development of robust automated wound analysis systems, particularly for chronic wounds in leprosy patients. This comprehensive dataset comprises 764 RGB images that reflect real-world clinical conditions, acquired over 1 year and 9 months (November 2021 to August 2023) using smartphone cameras. The images are organized into three distinct splits: 485 images for training, 122 for validation, and 157 unlabelled images reserved for testing (Figure 2).

Accompanying the training and validation images are detailed annotations that include detection bounding boxes and binary semantic segmentation masks in COCO format, which enable precise delineation of wound regions. Captured from 96 leprosy patients, of which 18 were female and 78 were male, with an average age of approximately 75.5 years (±9.9 years) and the dataset encapsulates the variability inherent in clinical image acquisition such as inconsistent illumination, varying noise levels, and contrast discrepancies. These features make CO2Wounds-V2 particularly valuable for developing algorithms that not only improve diagnostic accuracy but also exhibit strong generalization across diverse real-world conditions.

Table 1 provides a concise overview of the dataset distribution, allowing for further analysis with additional details. The

Table 1. Splitting Distribution.

Data Split	Total Images
Training Set	485
Validation Set	122
Testing Set	157

training and validation sets feature detection bounding boxes and binary semantic segmentation masks for each image, labeling each pixel as part of a wound or background. The testing set is provided without annotations for unbiased performance evaluation. The dataset is publicly available in Mendeley Data (https://data.mendeley.com/datasets/s2w7rjwz49/2), providing a rigorous benchmark for ongoing research in the field of medical image analysis.

Output Obtained

The provided figure illustrates the progressive impact of each module within the CO-WinF framework on chronic wound image analysis. Each row represents a distinct wound instance from the CO2Wounds-V2 dataset, with columns detailing different stages of processing. The first column shows the original wound image under natural smartphone-acquired illumination, often characterized by uneven lighting and poor boundary visibility. The second column displays the grayscale output post-illumination normalization and local contrast enhancement by the AINCE module, where wound textures become more pronounced and regional intensity homogeneity improves. Bounding boxes and outlines denote the region of interest extracted based on adaptive tile-wise remapping and edge-preserving smoothing. The third column presents the outcome of segmentation using the MEPS module, which integrates multi-resolution encoding with edge-aware attention to refine lesion boundaries. Here, the wound edges are accurately preserved, overcoming challenges posed by blurred margins. The overlayed purple masks represent the segmented wound regions. This visual comparison validates the effectiveness of AINCE in enhancing lesion detail and of MEPS in ensuring high-fidelity segmentation. The extracted and segmented regions serve as the basis for downstream classification using HSTFC, which integrates both spatial features and texture descriptors. Collectively, the image substantiates the contribution of each module in enhancing diagnostic reliability under real-world imaging conditions (Figure 3).

Performance of the Proposed Model

Figure 1, illustrates the performance of the proposed model during training and validation over 25 epochs. 1 (a) shows how both the training and validation accuracies steadily increase from the early epochs, with the model reaching above 90% accuracy by approximately the 10th epoch and continuing to improve with each subsequent iteration. This upward trend indicates that the model is rapidly learning representative features pertinent to wound segmentation, while the relatively close alignment between the two curves suggests that overfitting is minimized. 2 (b) presents the corresponding loss curves, where both the training and validation losses consistently decrease, reflecting effective convergence of the optimization process. The similar trajectories of these two loss curves further imply that the model maintains a healthy balance between fitting the training data and generalizing to unseen samples, underscoring the stability and robustness of the proposed approach (Table 2 and Figure 4).

The performance evaluation of the proposed model is summarized in Table 1, which delineates four key metrics Precision, Recall, F1 Score, and IoU Score across the training, validation, and testing phases. In the training phase, the model achieved slightly higher scores, indicating effective learning from the annotated data. The high Precision and Recall percentages during training of 96.12% and 96.50%, respectively reflect the model's robust ability to correctly identify the wound regions while minimizing both false positives and false negatives. The F1 Score of 95.20% confirms a good balance between these two aspects. A similarly high IoU Score of 89.00% in the training set demonstrates the model's accuracy in segmenting the wound regions with minimal overlap error.

When moving to the validation phase, the performance metrics show a minor decline, which is expected as the model is evaluated on unseen data. The precision reached 94.98% and recall 95.78%, with an F1 Score of 93.10% and an IoU Score of 87.09%. These numbers indicate that the model generalizes well while maintaining a high level of segmentation accuracy. In the testing phase, the metrics remain comparable to the validation results, with a precision of 94.50%, recall of 95.10%, F1 Score of 92.50%, and IoU Score of 86.50%. The slight drop in performance from training to testing confirms that the model is neither significantly overfitting nor under fitting, thereby ensuring its robustness and applicability in real-world scenarios. Overall, the consistent performance across training, validation, and testing phases validates the effectiveness and reliability of the proposed model for automated wound analysis.

Comparative Performance Analysis

The comparative performance table provides a comprehensive evaluation of various state-of-the-art techniques across multiple datasets, including HAM10000, CO2Wounds, the Brazilian leprosy dataset, and PAD-UFES-20. Each model's classification or segmentation ability is assessed using standard metrics such as accuracy, precision, recall, F1-score, and mean Intersection over Union (mIoU), highlighting the robustness and limitations

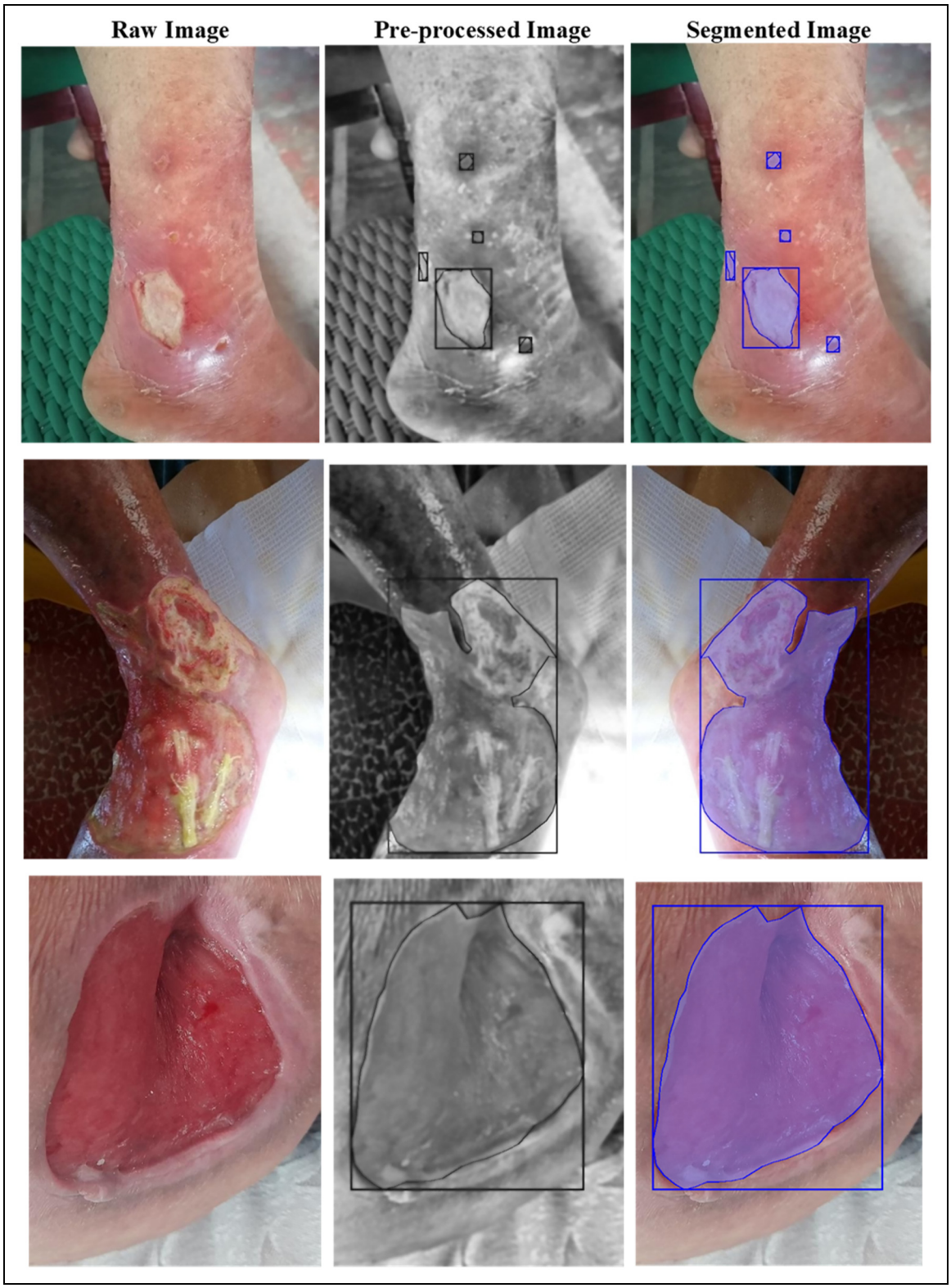


Figure 3. Output obtained by the proposed framework at different stages for three different samples.

Table 2. Performance of the Proposed Model in Terms of Precision, Recall, F1 Score and IoU Score for Training, Testing and Validation.

Performance Metric	Training (%)	Validation (%)	Testing (%)
Precision	96.12	94.98	94.50
Recall	96.50	95.78	95.10
F1 Score	95.20	93.10	92.50
IoU Score	89.00	87.09	86.50
Accuracy	98.38	97.53	97.20

of existing methods. For the CO2Wounds dataset, conventional U-Net architectures integrated with backbone networks like ResNeXt-50, ResNet101, EfficientNet, and Segformer achieved relatively high accuracy (~98.2-98.6%) but showed significant performance drops in precision (80.87-84.70%) and recall (77.80-81.99%), leading to suboptimal F1-scores (~76.61-79.44%) and low mIoU scores (66.96-70.13%). In contrast, the proposed model incorporating the MEPS segmentation module and HSTFC classification framework achieves a better balance, with a validation precision of 94.98%, recall of 95.78%, and F1-score of 93.10%, along with a notably improved mIoU of 87.09%. This demonstrates the proposed model's superiority in maintaining edge fidelity and accurate wound delineation, even under challenging conditions. Other datasets show mixed results: Vision Transformer (ViT) achieved strong performance on HAM10000 with high precision and recall, but deep models like ResNet and VGG showed inconsistency.

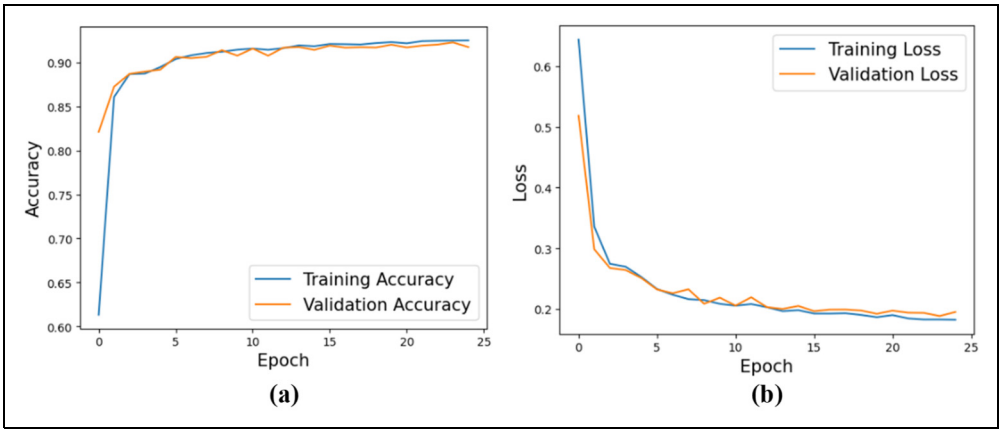


Figure 4. (a) accuracy and (b) loss plot obtained for the proposed model during training and validation over 25 epochs.

Reference	Dataset	Technique	Accuracy	Precision	Recall	F1-Score	mIoU
Sulthana et al[22]	HAM10000	S-MobileNet	93.2	95.58	93.56	95.58	–
Sanchez et al[23]	CO2Wounds	U-Net based ResNeXt-50	98.65	84.35	80.31	79.44	69.94
		U-Net based ResNet 101	98.28	80.87	81.10	76.61	66.96
		U-Net based EfficientNet	98.51	83.29	77.80	77.20	67.71
		U-Net based Segformer	98.59	84.70	81.99	79.26	70.13
Sanchez et al[23]	HAM10000	SVM	85.2	0.72	0.88	0.79	75
		KNN	81.2	0.69	0.65	0.66	70
		MobileNet	83	0.86	0.88	0.87	85
		ResNet 152 V2	50	0.43	0.46	0.45	75
		VGG16	79	0.75	0.74	0.74	59
		ViT	99	0.98	0.97	0.98	98
Remya et al[24]	Brazilian leprosy national r eferral open source dataset	ElasticNet Regression-LR	88	92	83	91	–
		XG Boost	78	87	72	82	–
		Random forest	80	90	72	86	–
Barbieri et al[25]	PAD-UFES-20	EfficientNetB3	76.00	81.00	82.00	74.00	
		Densenet121	71.60	66.82	63.65	64.85	
		VggNe t13	74.90	74.90	74.90	74.90	
		Mobilenetv2 + SVM	66.00	66.00	66.00	66.00	
		Vision transformer	81.95	82.33	81.95	81.51	69.34
Our proposed model	CO2Wounds	MEPS + HSTFC	97.53	94.98	95.78	93.10	87.09

Similarly, models tested on the PAD-UFES-20 and Brazilian leprosy datasets had moderate accuracy but lacked consistency in other metrics. Overall, the proposed method stands out for its well-rounded and robust performance, especially in segmentation-centric tasks under real-world conditions.

The present study was intentionally constrained to lepromatous wounds due to the availability and clinical validation of the CO2Wounds-V2 dataset, which provided a consistent ground truth for rigorous model training and benchmarking. At this stage, expanding to other wound types would require access to diverse, large-scale, and annotated datasets that reflect their heterogeneity. Therefore, while the framework is inherently extendable to other chronic wounds, the present validation remains specific to lepromatous ulcers.

Conclusion

The present research introduces a clinically aligned, computationally efficient framework for automated chronic wound analysis, designed to address the persistent challenges in boundary ambiguity, texture inconsistency, and class imbalance often encountered in wound segmentation and classification tasks. By integrating the Multi-scale Edge-Preserving Segmentation (MEPS) module with the Hybrid Spatial-Texture Fusion Classifier (HSTFC), the proposed approach achieves a harmonious balance between anatomical accuracy and diagnostic precision. The MEPS module adeptly retains intricate wound boundaries across varying scales, while the HSTFC strategically fuses deep spatial cues and handcrafted texture patterns through attention-guided mechanisms, culminating in a robust and discriminative feature representation. Empirical validation on the CO2Wounds dataset substantiates the method's clinical relevance, demonstrating superior performance with a validation precision of 94.98%, recall of 95.78%, F1-score of 93.10%, and a markedly enhanced mIoU of 87.09% signifying significant improvements over conventional U-Net derivatives and classification baselines. The architectural synergy of MEPS and HSTFC exemplifies how targeted innovation in preprocessing, segmentation, and classification can collectively elevate performance in medical image analysis. Thus, this study not only fills critical methodological voids but also sets a strong precedent for intelligent wound care systems, offering both theoretical advancement and tangible clinical value for leprosy treatment. As part of future work, this research can be extended by incorporating multi-wound datasets to evaluate cross-disease generalisation and fine-tune the architecture for broader clinical adoption.

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Ethical Approval

Institutional Review Board approval was not required.

Consent for Participate

All contributors agreed and given consent to participate.

Consent for Publication

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Author Contribution

The authors confirm contribution to the paper as follows and all authors reviewed the results and approved the final version of the manuscript.

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Data Availability

No data, models, or code were generated or used during the study

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